

Lyracist Pro Suite

Complete User Manual & Reference Guide

Lyracist Pro, LyracistDbEditor, and KSRotation Displays



Version 26.7 (July 2026 Release)

PAROLE Software Co.

1. Introduction to the Lyracist Pro Suite

Welcome to Lyracist Pro, the ultimate digital hosting and rotation management software suite tailored specifically for professional Karaoke Jockeys (KJs), DJ hosts, and entertainment venues. The suite is composed of three interconnected modules designed to coordinate lyrics playback, performer request flows, rotation scheduling, and public screen displays:

- Lyracist (Main App): The central desktop workspace where the host controls audio adjustments, video decoding, active singer rotations, catalog scanning, settings, and the mobile requests inbox.
- LyracistDbEditor: A standalone database utility used to browse, clean, import, and configure SQLite database files offline, ensuring no write conflicts or performance hits during live shows.
- KSRotation (Display Renderer): The projection renderer window that drives secondary and tertiary display monitors, rendering real-time lyrics overlay screens and custom themed rotation billboards.

-

- [IMAGE PLACEHOLDER: System Topology Setup]
- Graphic Details: A block diagram showing the main Lyracist hosting application running on a laptop, connected via HDMI to Monitor 2 (performer lyrics) and Monitor 3 (public rotation billboard), while local performer mobile phones connect via Wi-Fi to the local SignalR web server portal.

-

2. Lyracist (Main Host Application)

Lyracist runs as a high-performance desktop application built on modern WPF (.NET 10.0 Windows) architecture. It functions as the host's control dashboard, integrating music players, singer queue management, and live vocal DSP chains into a single interface.

2.1. The Main Dashboard & Playback Controls

The Live Deck sits at the top of the interface and controls playback of the current performer's track. It is driven by a high-performance LibVLC decoding engine to ensure maximum compatibility with video and CDG zip backdrops.

- Play / Pause Button: Starts or suspends the loaded song track. Can be triggered globally via the mapped Play/Pause hotkey.
 - Stop Button: Halts playback completely, resets the seek bar, and triggers ambient background or fill-in music automatically.
 - Seek Slider: Allows the host to click and drag to jump to any part of the active track.
-

- **Drive Status Indicator:** A glowing LED and text indicator on the main window title line showing whether library folders/drives are currently connected or missing. A green light (Drives Online) indicates all library folders are reachable, yellow indicates some folders are offline or no folders are configured, and red (Drives Offline) alerts you that the external storage drive is disconnected.
- **Volume Slider:** Adjusts the master volume output (0% to 100%) in real time.
- **Speed (Tempo) Slider:** Alters playback speed from 0.5x to 2.0x without shifting the audio pitch.
- **Pitch Transpose Slider:** Adjusts the key of the backing track in semitone steps from -6 to +6.

- [IMAGE PLACEHOLDER: Main Dashboard Controls]
- **Graphic Details:** A high-fidelity screenshot of the Lyracist main window, indicating the Play/Pause, Stop buttons, Seek slider, and deck speed/pitch controls highlighted with callout boxes.

2.2. Performer Queue & Rotation Management

The singer queue is built around a round-robin rotation algorithm to ensure that every singer gets a fair turn. The list displays active singers, their requested songs, transposition overrides, and status flags.

- **Adding Singers:** Enter the singer's name and select a song from the library, online Party Tyme stream, or custom external link (YouTube/Spotify).
- **Duet Partner Option:** An optional second singer slot. Enabling a duet prompts the system to configure dual audio profiles.
- **Done Singer Command:** Triggered when a singer finishes performing. It logs the performance to the singer's history, advances them to their next queued song, and moves them to the bottom of the list.
- **Reordering:** Drag and drop queue items to manually override the rotation order, or use the Move Up / Move Down buttons.

2.3. Dual-Singer Duet Audio & Vocal DSP Settings

When a duet is detected (a song is loaded with both a main singer and a duet partner), Lyracist Pro automatically instantiates individual DSP channels for both performers. The Vocal Settings dialog dynamically presents two configuration tabs: 'Main Performer' and 'Duet Partner'. Each profile allows independent control over:

- **Vocal Gain (dB):** Compels matching mic input volume amplification.
- **Transposition Preference:** Individual key transpose offset if singers require different key adjustments.
- **3-Band Parametric EQ:** Independent sliders to boost or cut Bass, Midrange, and Treble frequencies (+/- 10dB) to suit the vocal characteristics of each singer.
- **Dynamic Compressor Threshold:** Applies automatic volume leveling to manage peaks and whispers.

- Peak Limiter Threshold (dB): Caps maximum volume to protect the sound system against clipping and microphone drops.

-

- [IMAGE PLACEHOLDER: Duet Vocal Settings Window]
- Graphic Details: A screenshot of the Vocal Settings popup showing two tabs, 'Main Performer' and 'Duet Partner', with independent sliders for 3-band EQ, compressor, and peak limiter.

-

2.4. Venue Config & Customizable Transition Countdown

The Settings tab contains options to customize venue-specific branding and timing parameters:

- Venue Selector: Choose the current venue. Each venue saves its own local preferences and graphics folder path.
- Venue Slideshow Graphics: Add multiple graphic files (.jpg, .png) for each venue. When the rotation is idle, the public screen cycles through these graphics in a background slideshow loop.
- Auto-Advance Transition: Toggle the automatic transition countdown. When a singer completes a song, the system begins a visual countdown (customizable from 5s to 30s) before advancing to the next singer.
- Done Timer Overlay: Public display displays a ticking countdown bar. The KJ can click 'Advance Now' to bypass, or 'Cancel Countdown' to keep the screen paused.

2.5. Keyboard Shortcut Mapper (Learn Mode)

The 'Keyboard Hotkeys' tab allows the host to bind keyboard keys to main deck commands. To map a command, select the text field next to the action and press the desired key on the keyboard. Supported actions include Play/Pause, Stop, Done/Next Singer, Toggle Crawl Banner, Show Lyrics, and Show Rotation. The mapping system contains a built-in safety filter: hotkeys are automatically bypassed when the cursor is focused inside search fields or text inputs to prevent accidental triggers while typing names.

[IMAGE PLACEHOLDER: Keyboard Shortcut Learn Panel]

Graphic Details: A screenshot of the Settings - Keyboard Hotkeys tab showing input mapping boxes for Play/Pause, Stop, Done Singer, and Show Lyrics, with mapped keys like Space, Escape, and Return.

2.6. Product Registration Activation

Lyracist Pro features offline cryptographic license verification. In the Settings - Registration tab, the host enters their First Name, Last Name, Stage Name, Email Address, and Serial Key. The key is matched

against a SHA-256 + Custom Base-32 keygen algorithm. The validation state renders visual cards: a green checkmark (CheckmarkCircle24) indicates Lyracist Pro is fully activated; a red cross (DismissCircle24) represents validation failure; and a grey key represents awaiting inputs.

2.7. Integrated Mobile Request Portal & Wi-Fi Server

Lyracist Pro runs a private local web server via SignalR. Performers connected to the venue's local Wi-Fi can scan the QR code displayed on the screen to open the mobile web portal. Without needing internet access, performers can browse the host's local song catalog, view the current singer rotation queue, submit song requests directly to the host's Inbox, and spin the Scaryoke Wheel of Fortune to get a randomly assigned category and song.

Performer requests are split into Karaoke requests (singer performing) and background Music requests. A separate 'Search Music' tab is provided on the mobile portal for browsing the background music collection. In addition, performers can tap reaction buttons (🍷 , 🔥 , ❤️ , 🍀 , 🎉 , 🏆) to send live, animated, color emoji reactions that drift and fade across both projection screens in real time.

2.8. LYRX Parity Visual & Audio Enhancements

We integrated two major performance features from LYRX Karaoke: AI Vocal Suppression ('Kill Vocal') and the Audio-Responsive Visualizer. The AI Vocal Suppression uses native LibVLC EQ mid-band cancellation (500Hz to 4kHz) to suppress lead vocal tracks by -20dB, transforming normal audio/video files into instant backing tracks. The Audio-Responsive Visualizer renders neon-gradient spectrum bars that react in real time at the bottom of the public lyrics window when playing, or slow ocean waves when idle.

[IMAGE PLACEHOLDER: Live Deck with Kill Vocal Toggle]

Graphic Details: A screenshot of the deck playback panel highlighting the 'Kill Vocal (AI)' toggle switch situated next to the 'Reset Audio' button.

2.9. Audio Outputs & Hardware Mixer Mode

Lyracist Pro supports routing audio streams to separate physical output channels on external hardware mixers (e.g., Pyle PMXU88BT).

- Karaoke Audio Output: The output channel dedicated to playing the singer's backing track, configured in settings.

- Background Music (BGM) Output: The output channel dedicated to ambient fill-in, opening, and end-of-rotation playlists, keeping background music independent of the performer's microphone channel.
- Hardware Mixer Mode: Bypasses all software DSP volume, EQ, compressor, and limiter parameters, sending a clean, flat 100% volume signal directly to the external physical mixer to avoid muddy sound from double-processing.

2.10. Performer XP, Levels & Progression

The suite features a built-in gamification system for performer engagement. Experience points (XP) are calculated as: $XP = \text{Total Songs Sung} * 100 + \text{Average Performance Score}$. Singers automatically unlock performance titles (e.g., Shower Singer, Pub Regular, Vocal Powerhouse, Karaoke Legend) and achievement badges (Debut, Legend, Rising Star, Crowd Pleaser, High Scorer) visible on the rotation billboard and tablet portal leaderboards.

3. LyracistDbEditor (Offline Catalog Manager)

LyracistDbEditor is a utility designed to manage the SQLite databases offline, avoiding locks or interference while the main show is running.

- Library Importer: Import directories of MP3, WAV, MP4, and zipped MP3+G files. It automatically extracts song titles and artist names from file names based on customizable templates (e.g. 'Artist - Title' or 'Title - Artist').
- Singer Database: View and edit all performer records. You can override completed song counts, last-sang timestamps, and clear ratings.
- Slideshow Setup: Map and verify venue graphic slideshow paths to ensure they match local directories.
- Database Purging & Backups: Create compressed database backups before major show events, or purge old history entries.

-

- [IMAGE PLACEHOLDER: LyracistDbEditor Catalog Panel]
- Graphic Details: A screenshot of the database editor showing the song library grid, search filters, and import folder directory settings.

-

4. KSRotation (Public Display Projection)

KSRotation controls the secondary and tertiary video outputs. It renders lyrics and rotates the performer billboard display modes.

- Normal Queue Mode: Renders a modern, clean dark-themed list of upcoming performers.
- Star Wars Crawl: Renders the performer queue as a 3D perspective scrolling text crawl, scrolling into deep space with customizable intro banners ('A long time ago, at a karaoke bar far away...').
- Vegas Marquee Mode: Showcases the current and next performer names inside an animated, glowing neon lightbulb marquee grid.
- Vinyl Turntable Mode: Renders a spinning record sleeve graphics template with track details.
- Transition Auto-Advance Countdown: Displays a progress bar and remaining seconds ticker on the public screen when a song finishes, letting performers know when the next singer is about to launch.

-

- [IMAGE PLACEHOLDER: Star Wars Crawl Display Mode]
- Graphic Details: A rendering of the public projection screen showing the performer queue scrolling as a 3D Star Wars text crawl into space, complete with background stars.

-

-

Section: Database Manager / Catalog Book Exporter

[Update Details]

Operators can now export the entire song database (Karaoke or Music) into Word (.docx) or PDF format directly from the Lyracist Database Manager.

1. Exporter Panel:

- Located in the right pane of the Lyracist Database Manager under "Catalog Book Generator".
- Provides two dropdown selectors:
 - Catalog: Choose between the "Karaoke Catalog" (tracks marked as karaoke) and the "Music Catalog" (standard audio tracks).
 - Format: Choose between "Word Document (.docx)" and "PDF Document (.pdf)".

2. Generated Fields & Formatting:

- The generated books are structured as clean, professionally styled tables showing:
 - Artist: Alphabetically ordered performer names.
 - Title: Song titles.
 - Length: Formatted track durations (MM:SS).
 - Genre: The embedded audio genre (automatically populated during metadata scans).

3. Pagination & Output:

- The document is fully paginated, displaying "Page X" or "Page X of Y" in the bottom footer.
- For Word (.docx) books, a repeating header row is defined so column titles automatically reappear on the top of each page.
- When generation completes, Lyracist automatically launches the default system viewer to open and print the document, which is saved in the "Reports" directory.

Section: Display Projection Window / DJ Banner Screen & Selection

[Update Details]

Operators can now configure and display custom DJ branding, logos, and promotional banners on a designated projection screen.

1. DJ Banner Screen Target:

- Available in both Lyracast (Settings page & Karaoke page) and KSRotation (Settings tab).
- Allows operators to choose which connected monitor will project the borderless, full-screen DJ banner.
- When enabled/opened, it launches the screen on the target monitor and keeps it topmost.

2. Upload Banners:

- Operators can upload multiple banner images (PNG, JPG, JPEG, GIF, BMP) from their system via the 'Upload' button.
- Files are copied into a local DJBanners settings subdirectory automatically.

3. Select Active Banner:

- If more than one banner is uploaded, operators can select which banner is active and displayed from a dropdown list of available images.
- Selecting a different banner updates the projected screen instantly without reloading the window.

Section: Display Projection Window / Casting Support

[Update Details]

Operators can now select different casting targets for the Singer Rotation Billboard screen in both Lyracist and KSRotation applications, in addition to standard physical monitors.

1. Casting Target Dropdown:

- Located in the "Settings" tab under "Displays & Projection" (or "Display & Projection" in KSRotation).
- Dropdown options include Monitor, Miracast, Chromecast, Browser Cast, AirPlay, and Wireless HDMI.
- When a casting target is selected, the application (Lyracist or KSRotation) initializes the appropriate casting sender/server in the background.

2. Off-Screen Rendering:

- Selecting a casting target (such as Chromecast or Browser Cast) creates and renders the Rotation Window / Singer Display Window in an off-screen buffer (invisible to the host desktop) to capture frames dynamically.
- This ensures the KJ laptop screen is not cluttered while casting frames.

3. Browser Cast Server:

- The self-hosted Browser Cast option spins up a local web server (<http://localhost:8080/rotation/>) to allow any browser on the local network to view the singer rotation billboard in real-time.

Section: Database Manager / Catalog Book Exporter & Failed Artist List

[Update Details]

Updates have been made to the Catalog Book Exporter formats and the Slow Metadata Extractor.

1. Supported Formats and Sorting:

- The Catalog Book Exporter now supports exporting the catalog as a plain text (.txt) file.
- The format selection dropdown has been sorted in the following order: PDF Document (.pdf), Text File (.txt), and Word Document (.docx).

2. Metadata Scan Failed Artist Export:

- When running a Slow Metadata Scan for songs with unknown artists, any files that were unable to be resolved and updated with a valid artist during the scan are tracked.
- Upon completion or cancellation of the scan, a plain text file listing all unresolved file paths is automatically exported to the "Reports" directory, named "Failed_Artist_Updates_[Timestamp].txt".

Section: Integration Settings / Party Tyme Removal

[Update Details]

The Party Tyme karaoke streaming integration has been removed from the application.

1. Removal of Streaming Tab:

- The "Party Tyme Online" tab has been removed from the main Karaoke dashboard view.
- The "Party Tyme (Streaming)" tab has been removed from the Special Occasions Music search dialog.

2. Settings Panel Clean-up:

- The subscription fields (Client ID and Client Secret) for Party Tyme have been removed from the Settings panel under APIs & Logins.

3. Queue & Badge Updates:

- The "Party Tyme" filter checkbox and visual badge triggers have been removed from the singer rotation queue list.

Section: Performer Profiles, Song Tagging & Singer Avatars

[Update Details]

A comprehensive performer profile system and song tagging support have been integrated into Lyracist.

1. Performer Profiles & Login:

- Patrons accessing the Patron Request Portal can log in or register using their Performer Name and a 4-digit PIN.
- Profile settings include email registration, custom vocal range specification (e.g. Tenor, Soprano), and custom title assignment (e.g. The Screamer).

2. Singer Avatars & Custom Uploads:

- Performers can customize their profile avatar using a dropdown selector.
- Supports default silhouette, Gravatar logo generation (via email MD5 hash), and direct mobile photo/selfie uploads (capped at 2MB).
- Avatars are displayed as circular icons in the rotation board on both the mobile Patron Portal and host WPF Singer Queue ListBox.

3. Database Tagging:

- Songs can now be tagged with comma-separated categories to organize, filter, and challenge performers.

// Edited on Aug 19, 2026 @ 11:24:00 -> Added Help & About tabs in LyracistTrivia and timer override documentation

Rotation Management & Round Cycle Tracking (` Lyracist` , ` KSRotation` , & ` KSRotation.Maui`) - (August 2026)

Section: Performance & Rotation Operations / Singer Queue Management

[System Details]

1. Float Current Singer to Top of Rotation:

- ****User Interface Controls****:

* `Lyracist` : Checkbox on the ****Rotation**** page queue toolbar, ****Karaoke**** page queue bar, and persistent toggle under ****Settings -> Monitors & Screen Assignments****.

* `KSRotation` (WPF): Checkbox on the main rotation toolbar with automatic settings persistence.

* `KSRotation.Maui` : Checkbox in the Active Rotation Queue header bar.

- ****Queue Behavior****:

* When enabled, the performer currently at the microphone automatically floats to the very top (index 0) of the list.

* Upon song completion (`Done` / `✓`), the finished performer is moved to the back of the active queue and the next active performer automatically floats to the top (index 0).

* Eliminates the need for the DJ/KJ to scroll down long lists during active shows.

2. 1st Singer in Rotation (Round Start Anchor Flag):

- ****Visual Indicator****:

* A red  1ST` badge is displayed next to the singer designated as the first performer of the rotation round, and that singer's row is also highlighted with a red background (alongside the existing yellow "now performing" and blue "up next" row highlights), so the 1st singer stands out at a glance.

* When the 1st singer returns to index 0 / current status, the DJ knows a complete rotation cycle has been completed.

- ****Manual Re-anchoring & Clearing****:

* Right-clicking any singer and selecting `Set as 1st Singer (Round Start)` or clicking the  button reassigns the anchor to that singer, clearing it from previous holders.

* Clicking the same action again on the singer who already holds the flag (the button turns red / the menu item changes to `Clear 1st Singer Badge`) undoes the mistake — it hands the flag to the next active singer in rotation order rather than leaving nobody flagged, so an accidental click never needs a specific replacement singer picked.

* When new singers are added, the system automatically ensures exactly one active singer holds the start flag.

- **Cross-App Sync**:

* The `isRotationStart`` flag is synchronized via the REST/WebSocket `/api/rotation`` feed between `KSRotation`` and `Lyracist``.

Lyracist Live Trivia (`Lyracist``, `KSRotation``, & `Lyracist.Trivia``) - Integrated Game Master & Intermission System (August 2026)

Section: Entertainment Modules / Lyracist.Trivia & Pub Trivia Game Master

[System Details]

1. Multi-Monitor Big Screen Projection & Auto-Casting in `KSRotation``:

- **" Open Big Screen" Button**:

* Directly on the ** Trivia Game Control`** tab in `KSRotation``.

* Toggles and focuses the full-screen 16:9 venue projection window (`TriviaDisplayWindow``) on the selected target monitor.

- **Automatic Projection on Game Start**:

* Clicking **"▶ Start Game"** in `KSRotation`` automatically opens and positions the trivia projection window onto the designated secondary monitor/TV so patrons can immediately see questions, timer countdowns, and options.

- **Dynamic Repositioning**:

* If the target monitor is switched in display settings, `TriviaDisplayWindow`` dynamically relocates to the newly selected monitor.

2. Hybrid Game Engine Architecture & Integrated Host Decks:

- **Lyracist` Application Integration**:

* Top-level navigation items for **Trivia** (interactive host deck) and **Trivia Settings** (3-column no-scroll dashboard for timers, points, bonuses, and Wi-Fi credentials).

- **KSRotation Application Integration**:

* Dedicated main window tabs for **Trivia** and **Trivia Settings**.

- **Deduplicated Venue & Game Master / DJ Settings**:

* Trivia settings in both apps pull directly from existing Venue (`AppSettings.SelectedVenue / MainViewModel.VenueName`) and DJ / Host (`AppSettings.DjName / MainViewModel.DjName`) properties.

* Duplicate textboxes were removed and replaced with clean synchronized information badges.

- **3-Column No-Scroll Trivia Settings Dashboard**:

* Divided into 3 horizontal columns:

- Column 0: Question Timers & Flow, Scoring & Bonus Multipliers (Base Points, Wrong Answer Deduction Points [Default: 0], Speed Bonus, Streak Multipliers)

- Column 1: Active Venue & Host Information Badge, Intermission & Pre-Game Countdown

- Column 2: Local Network & Wi-Fi Access, Save / Reset Action Cards

* Eliminates vertical scrolling on standard desktop resolutions.

- **Automated Game Pause & Resume Synchronization**:

* Automatically pauses the trivia round when DJ Banners, Special Events, or Rotation screens take over the projection display.

* Displays `"Game Paused: {Reason}"` on the host console and `#pause-banner` on mobile buzzers, safely freezing timers and blocking submissions until the overlay is closed.

- Run as a standalone application (`Lyracist.Trivia`) for dedicated venue Trivia Nights, or trigger rapid fill-in / intermission rounds directly during karaoke shows.

- Shared core library `Lyracist.Trivia.Core` handles SQLite persistence, real-time scoring, speed bonuses, streak multipliers, and player socket communication.

- **Venue & Game Master Branding Configuration**:

* Configure Venue Name (e.g., "The Neon Lounge") and Game Master / Host Name (e.g., "Trivia Master") on the **  Settings & Display** tab.

* Updates dynamically across all projection screens, scoreboard banners, and mobile web buzzers in real time.

- **Multi-Monitor Display Resolution & Placement**:

* Select any connected monitor (Monitor 1, Monitor 2, HDMI TV, or projector) from the Game Master console.

* Uses DPI-aware screen bounds positioning without forcing to Monitor 0.

- **Projection Screen Escape & Close Controls (`TriviaDisplayWindow`)**:

* **Escape Key (`Esc`)**: Pressing Escape immediately closes and exits the projection display window.

* **Floating Close Button (`X`)**: Sleek translucent circular close button in the top-right corner, illuminating red on hover for instant 1-click closure.

* **Right-Click Context Menu**: Right-clicking anywhere on the screen displays quick options to close (`X Close Projection (Esc)`) or toggle fullscreen (`⏏ Toggle Fullscreen (F11)`).

* **Double-Click & Drag**: Double-clicking the display toggles between windowed and fullscreen maximized modes, with click-and-drag window repositioning.

- **Unified 16:9 Pre-Game Lobby & Category Showcase Screen (`TriviaDisplayWindow`)**:

* Combines all pre-game elements into a clean, modern split layout:

1. **Left Hero Column (70% width)**: Full-featured 16:9 Category Announcement Banner (`TriviaData/Banners/{pack}.png`) with ambient glow, tonight's category theme title, and topic subtitle. Cross-fades dynamically when the host selects a different pack in the console dropdown.

2. **Right Onboarding Stack (30% width)**:

- **  Game Start Countdown Clock**: Large digital timer with pulsing badge (`⌚ GAME STARTS IN: mm:ss`) that counts down to launch.

- **  1. Connect to Wi-Fi Card**: Dedicated scan-to-connect Wi-Fi QR code (`WifiQrCodeImage`), SSID name, and password.

- ** 🎲 2. Join Trivia Game Card **: Dedicated scan-to-join mobile buzzer QR code (` QrCodeImage `), direct URL, and buzzer onboarding instructions.

3. **Bottom Connection & Copyright Bar **: Displays mobile buzzer URL (` 🎲 PLAY ON YOUR PHONE: ... `), company copyright information (` © 2026 PAROLE Software - All rights reserved. `), and app branding across all projection screens.

4. **Continuous Scrolling Marquee **: Bottom/top ticker bar displaying venue branding, Game Master name, game rules, and buzzer tips.

* ** 🎯 Launch Pre-Game Lobby & Countdown **:

- A single primary action button in the Game Master console that opens/focuses the big screen, locks the category banner, starts the pre-game countdown, and activates the lobby.

- Accompanied by ` "▶ Start Game Now (Skip Countdown)" ` for immediate question launch.

- **Top Scrolling Player & Score Marquee **:

* Continuous horizontal marquee ticker running along the top of the venue display screen (` TriviaDisplayWindow `).

* Shows all registered players and teams with their live point totals.

* The leading top scorer is dynamically highlighted with a distinctive gold crown badge (` "👑 1. Player (Team): 3,450 pts" `).

- **3-Tab Game Master Console & Settings Page **:

* ** 🎯 Live Game Master Tab **: Streamlined 3-column live command deck with question pack selection, auto-run switch, live question & 2x2 options visualizer, QR connection code, live scoreboard, and round flow controls.

* ** ⚙️ Settings & Display Tab **: Full configuration page for Intermission delay (configurable minutes), auto-start toggle, display monitor selector, question timers (15s), wrong answer elimination speed (5s), post-reveal buffer (5s), base points, speed bonus toggle, sound effects, server port, venue branding, and persistent "Save All Settings" button.

* ** 👥 Players & Teams Tab **: Roster grid showing player names, team names, total points, streaks, correct answer counts, total answered, and individual player kick/removal controls.

- **Post-Game Intermission Countdown & Automatic Next Game Start**:

* When a game finishes, if auto-start is enabled, an intermission countdown runs for the configured duration (e.g. 3 minutes).

* Venue display, GM console, and mobile web app display a live ticking countdown banner (` " 🎮 NEXT TRIVIA ROUND STARTS IN: mm:ss" `).

* Host can click ` "▶ Start Now (Skip)" ` to bypass the intermission, and once expired, the engine automatically selects the next question pack (or reshuffles) and launches the new round.

- **Connect Instruction Screen & Dedicated Trivia Navigation in ` Lyracist ` and ` KSRotation `**:

* ` ` Lyracist ` `: Access ` "👁️ Preview Connect Screen" ` in ` SettingsPage ` (Column 4: Connect & Request Instructions), with full Trivia controls and rules configuration residing in dedicated **Trivia** and **Trivia Settings** navigation pages.

* ` ` KSRotation ` `: Preview the Wi-Fi and Request Connect screen directly from ` Connect & Request Instructions ` under the **Display** settings tab, while Trivia rounds and configuration remain cleanly dedicated to the **Trivia** and **Trivia Settings** tabs without toolbar clutter.

- **Configurable Questions Per Game**: Set the number of questions per game round (5, 10, 15, 20, 25, 50, 100, or custom editable value in the host console) stored in ` trivia\_settings.json ` . Slices the shuffled question pack to the exact number of desired questions.

- **Game Complete Winner Celebration & Team Announcements**:

* Upon completing all allotted questions, the game transitions to a celebration screen on venue big screens (` TriviaDisplayWindow `) and player mobile devices (` trivia.html `).

* Announces the winning champion and final score.

* If the winner is a team, prominently displays ` "👑 WINNING TEAM: {TeamName}" ` and lists all players on that team (` "Team Players: Alice, Bob, Charlie" `).

* Mobile phones display the winning team/player, team member list, the player's personal placement and final score, and the complete top 10 leaderboard.

- **Zero-Config LAN Mobile Web Server (` TcpListener `)**: Direct socket HTTP server listening on ` 0.0.0.0 ` (all interfaces) with RFC 1918 LAN IP resolution, allowing

smartphones and tablets on local Wi-Fi to load buzzers and buzz in without Windows URL reservation restrictions or admin privileges.

- ****Dynamic Question Shuffling & 2,100 Total Questions****: When any category is selected or played, the entire question database (150 curated questions per pack across all 14 categories, totaling 2,100 questions) is dynamically shuffled, ensuring questions appear in a fresh, non-repeating order each time a category is chosen.

- ****Target Monitor Selector****: Host console features a monitor selection dropdown (Monitor 1, Monitor 2, TV/HDMI) allowing dynamic window positioning and DPI-aware full-screen projection onto any connected venue monitor.

- ****Auto-Running Game Loop****:

- * ****15-Second Answering Window****: When a question begins, players have 15 seconds to buzz in on their smartphones or venue screens.

- * ****5-Second Sequential Wrong Answer Elimination****: When the 15-second window expires, incorrect answers progressively fade out one every 5 seconds until only the correct answer is left standing and illuminated.

- * ****5-Second Post-Reveal Buffer & Progression****: Once only the correct answer is illuminated with full explanation and point calculations, the game waits 5 seconds and then automatically advances to the next question.

2. Centralized Trivia Storage (`TriviaData/`):

- Located directly in the solution root at `TriviaData/`.

- Contains SQLite database `trivia.db`, host settings `trivia\_settings.json`, and question pack directory `TriviaData/packs/`.

- ****15 Comprehensive 150-Question Category Databases (2,250 Questions Total, No Duplicates)****:

- * ****Famous Lines & Sayings From Movies**** (`famous\_movie\_quotes.json`): 150 questions covering AFI Top 100 movie quotes, iconic catchphrases, and unforgettable dialogue across Hollywood history.

- * ****Bikers & Motorcycles**** (`biker\_trivia.json`): 150 questions covering Harley-Davidson engines, Indian heritage, European/Japanese classics, rallies, and chopper mechanical lore.

* **Rock & Roll** (`rock\_and\_roll.json`): 150 questions covering classic rock, 70s/80s bands, albums, and guitar legends.

* **Country Music** (`country\_music.json`): 150 questions covering outlaw country, 90s hits, and Opry legends.

* **Geography** (`geography.json`): 150 questions on continents, oceans, mountains, rivers, and landmarks.

* **State Capitals** (`state\_capitals.json`): 150 questions covering all 50 US State Capitals + world capitals.

* **History** (`history.json`): 150 questions on world & US history, ancient civilizations, and major milestones.

* **Complete the Lyric** (`complete\_the\_lyric.json`): 150 singalong lyric questions across rock, pop, and karaoke favorites.

* **TV Shows** (`tv\_shows.json`): 150 questions covering classic sitcoms, 90s hits, drama series, and cartoons.

* **Sports** (`sports.json`): 150 questions covering NFL, MLB, NBA, NHL, soccer, Olympics, and golf.

* **Logos & Brand Slogans** (`logos\_and\_slogans.json`): 150 questions covering advertising taglines, mascots, and brand history.

* **Music & Karaoke Legends** (`music\_legends.json`): 150 questions celebrating iconic vocalists, chart-topping legends, and karaoke hall-of-fame artists.

* **80s & 90s Pop Culture** (`pop\_culture\_80s\_90s.json`): 150 questions on retro toys, video game classics, 80s/90s movies, and nostalgic trends.

* **Blockbuster Movie Soundtracks** (`movie\_soundtracks.json`): 150 questions covering blockbuster film scores, Oscar-winning theme songs, and needle-drops.

* **Pub Trivia All-Stars** (`pub\_general\_knowledge.json`): 150 questions spanning science, literature, food & drink, myths, idioms, and pub favorites.

3. Mobile Player Portal (`trivia.html`):

- Zero-installation mobile web app hosted directly by the embedded web server (default port 8085).

- Patrons scan an on-screen QR code on their smartphones to register their player/team name and buzz in using 4 large, responsive touch cards (A, B, C, D) with haptic feedback.
- Live rank indicators, streak counters, and real-time progressive elimination fading on phone buzzers.

4. 16:9 Venue Display Projection (`TriviaDisplayWindow.xaml`):

- High-fidelity full-screen monitor output featuring dynamic category headers, large question prompts, 4-color option cards with sequential opacity elimination fading, configurable 3-second warning timer badges, and live team leaderboard podiums.

// Edited on Aug 17, 2026 @ 12:31:30 -> Document single-song duet auto-clearing and session history persistence

Lyracist Pro & KSRotation Update - Duet / Backup Partner Support (August 2026)

Section: Singer Request Board & Rotation Management / Remote DJ & Mobile Portals

[Update Details]

1. Single-Song Duet Auto-Clearing & Session Reporting:

- When a performer finishes their song (`FinishSingerSong` or `AdvanceToNextSinger`), the duet partner is automatically archived into the session performance history database and then cleared from the active queue for their subsequent rounds.
- Preserves historical records for the entire night in `SongPerformance` (KSRotation), `PerformedSong` (Lyracist), Night Database snapshots, CSV exports (`Duet Partner` column), and PDF summary reports (`(with {PartnerName})`).

2. Mobile Patron Request Portal (`PatronPortal.html`):

- Added an optional "Duet / Backup Partner (Optional)" input field to the song submission form.

- Automatically displays for Karaoke requests and hides when Music (track playback) is selected.

- Forwards the partner's name as ``duetPartner`` with the request JSON payload.

- Rotation list displays ``(with {PartnerName})`` beside performer names.

3. Remote DJ Web Board (``dj.html``):

- Added "Duet / Backup Partner (Optional)" input field to the "Add Performer to Rotation" form.

- Added ``(with {PartnerName})`` badges to both the live queue cards and incoming request cards.

4. KSRotation Desktop (``MainWindow.xaml``):

- Added a dedicated "Duet Partner" editable column in the rotation queue grid.

- Incoming requests sidebar shows ``(with {PartnerName})`` for quick verification before accepting.

5. KSRotation.Maui (.NET MAUI):

- Added Duet Partner entry to the top Add Performer bar.

- Added Duet Partner field to the Edit Performer dialog.

- Displays ``(with {PartnerName})`` on rotation items and incoming requests.

6. Billboard Projection (``DisplayViewModel.cs``):

- Formats performer names as ``{Singer} & {Partner}`` on "Now Performing", "Up Next", and rotation crawls.

// Edited on Aug 17, 2026 @ 11:22:00 -> Document DJ QR Code enlarged popout modal and optical contrast enhancements

Lyracist Pro & KSRotation Update - DJ QR Code Enlarged Popout & Optical Contrast (August 2026)

Section: Singer Request Board / DJ Remote Control Portal

[Update Details]

1. Enlarged DJ Control QR Code Popout Modal:

- Clicking the DJ Control QR code on the main request panel opens a dedicated popout dialog (``DjQrCodePopoutWindow``).
- Features an extra-large (280x280) high-resolution QR code rendering with ``NearestNeighbor`` pixel scaling for easy optical scanning across the room.
- Prominently displays the active DJ Security PIN in large 24pt bold red font with a one-click "Copy PIN" button.
- Displays the full DJ web portal address with a one-click "Copy URL" button.
- Targeted Interaction: Popout activates only when the DJ Control Portal mode is active (``IsDjQrVisible``), leaving the singer song request QR code unaffected.

2. High-Contrast QR Code Optical Scanning:

- Standardized DJ and Wi-Fi QR codes to pure black on white (``[0, 0, 0]`` on ``[255, 255, 255]``) with a 21:1 contrast ratio, ensuring instant scanning on older tablet cameras (such as iPad Air).
- Added a red accent border (``#EF4444`` , 2.5px) around the request panel container when DJ mode is active to clearly distinguish DJ mode.

// Edited on Aug 17, 2026 @ 09:55:00 -> Document Last Song priority banner display on all non-lyric screens

Lyracist Pro Update - Priority "Last Song" Banner on All Non-Lyric Screens (August 2026)

Section: Display & DJ Banners / Special Event Banners

[Update Details]

Added priority display handling for the "Last Song" event banner (`LastSong.png`):

- Highest Banner Precedence: When the "Last Song" event banner is selected or activated (via Settings, Karaoke Controls, or Remote DJ Board), it takes immediate priority over all other DJ and party banners.
- Multi-Screen Projection: The Last Song banner automatically projects on all non-lyric projection displays (the DJ Banner Screen and the Singer Rotation Billboard Screen), ensuring the entire venue sees the finale announcement.
- Uninterrupted Lyrics: The dedicated Lyric Projection screen continues rendering transparent CDG/scrolling lyrics without disruption so the final performer can complete their performance.

// Edited on Aug 15, 2026 @ 10:20:00 -> Document Session Performed Songs List & Textbox with 5-color rotation in Rotation Page

Lyracist Pro Update - Session Performed Songs History in Singer Rotation (August 2026)

Section: Singer Rotation / Session Performance History

[Update Details]

Added a dedicated Session Performed Songs column (Column 2) to the Singer Rotation page:

- Live Performance Tracking: Automatically records every completed song performance during the session, including performer name, song title, artist, key transposition, and timestamp.
- Dual Display Modes:

1. 5-Color Rotating List: Each performed song entry rotates sequentially through 5 distinct theme-tailored colors (Violet, Cyan, Emerald, Amber, Rose) in both Dark and Light color schemes for high visual clarity.

2. Session Textbox: Provides a clean multi-line read-only TextBox containing the complete text log of performed songs.

- Session Management: Includes 'Copy Session Songs to Clipboard' and 'Clear Session Performed Songs' quick-action buttons.

// Edited on Aug 15, 2026 @ 10:00:00 -> Document startup splash window bypass default
Lyracist Pro Update - Bypass Startup Splash Window (August 2026)

Section: Application Options / Startup Behavior

[Update Details]

The startup splash window pop-up has been disabled by default (` ShowSplashOnStartup = false `). Lyracist now launches directly into the main application window without displaying a splash screen or artificial loading delays.

Lyracist Pro Update - Title Bar Venue & DJ Info Block (August 2026)

Section: Main Application Window / Title Bar Integration

[Update Details]

Added dynamic Venue and DJ name indicators directly to the main window Title Bar:

- Placement: Positioned after the application title and before the pending request indicator bulbs.

- Stacked Format:

- Top Line: `Venue: <name>` (bound to `VenueTitleText`, updating live when active venue changes in Settings).

- Bottom Line: `DJ: <name>` (bound to `DjTitleText`, updating live when active DJ name changes in Settings).

Lyracist Pro Update - Settings Display Tab Controls & QR Code Overlay (August 2026)

Section: Settings Page / Display Tab / QR Code & Screen Controls

[Update Details]

Added explicit screen activation toggles and request QR Code overlay controls on the Display tab:

1. Screen Activation Checkboxes:

- "Enable DJ Banner Screen": Toggles the DJ Banner projection window on/off (``IsDjBannerActive``).

- "Enable Lyrics Projection Screen": Toggles the Lyrics projection window on/off (``IsLyricsActive``).

- "Enable Singer Rotation Billboard Screen": Toggles the Rotation billboard projection window on/off (``IsRotationActive``).

2. Request QR Code Overlay Controls:

- "Show request QR Code overlay on DJ Banner": Toggles rendering the patron request portal QR code overlay directly onto active DJ Banners (``IsDjBannerQrCodeEnabled``).

- "Connect Instructions & QR Code Screen": Selects the target display screen/monitor for projected Wi-Fi and Song Request dual QR code banners (``ConnectInstructionsScreen``).

Lyracist Pro Update - Settings Display Tab (August 2026)

Section: Settings Page / Display Tab (4 Equal Columns)

[Update Details]

A dedicated "Display" tab has been added to the Settings screen in Lyracist. Displays & projection monitor assignments, DJ banner images, special event banner mappings, Star Wars crawl & spaceship overlay settings, and Connect & Request Instructions have been consolidated onto the Display screen.

1. 4 Equal Columns Grid Setup (referencing KsRotation layout):

- Column 1: Monitors & Screen Assignments (Lyrics Projection Screen, Singer Rotation Billboard Screen, DJ Banner Screen, Billboard View Mode, Mirror Lyrics).
- Column 2: DJ & Event Banners (DJ Banner Image upload/select/delete, Special Event Banners Mapping & Save button).
- Column 3: Star Wars Crawl & Spaceship Overlay (Crawl Intro Template & preview, Custom Crawl Text, Spaceship Overlay Font Size, Duration, Frequency, Snippets List & controls).
- Column 4: Connect & Request Instructions (Wi-Fi Password, Target Screen selector, Dynamic Banner Status info).

Lyracist Pro & KsRotation - Performer Portal Update (August 2026)

Section: Connect & Request Instructions Screen / Wi-Fi Password Manager

[Update Details]

A dedicated Connect & Request Instructions feature has been added to display scan-to-connect Wi-Fi and song request QR codes directly on projection monitors.

1. Scan-to-Connect Wi-Fi & Song Request Dual QR Code Banner:

- Automatically detects active Wi-Fi network SSID and song request portal URL.
- Generates a dynamic vector graphic (`ConnectInstructions.png`) featuring a WPA/WPA2/NoPass Wi-Fi scan-to-connect QR code (forest green `#064E3B`) and a song request portal QR code (pure black `#000000`).
- Clearly displays Wi-Fi network SSID, password status ("No Password Required" or explicit password), and portal URL.

2. Dynamic Target Screen Resolution Detection:

- Automatically detects the physical pixel dimensions (`width` x `height`) of the target display monitor (`1080p` , `1440p` , `4K 3840x2160` , etc.).
- Dynamically scales vector paths, fonts, card bounds, and QR code module density (`pixelsPerModule`) to guarantee razor-sharp 1:1 pixel rendering on any screen size.

3. Persistent Wi-Fi Password Store (`wifi\_passwords.json`):

- Automatically remembers Wi-Fi passwords entered for specific SSIDs (venue Wi-Fi, travel router, mobile hotspot).
- Upon detecting a connected Wi-Fi network SSID, the application automatically recalls and populates the saved password into the Wi-Fi password input box.
- Entering or updating a password automatically updates `wifi\_passwords.json` for that SSID.

4. Dedicated Green GroupBox Layout & Screen Selector:

- Located in the Display settings tab inside a dedicated Green GroupBox styled under Column 1 (directly under Display & Projection).

- Features a dedicated ****Target Screen / Monitor**** selection dropdown with options: "None", "All Screens / Monitors", or individual connected display monitors.

5. Integrated Help System Topic Expansion:

- Added dedicated 'Connect & Wi-Fi Instructions' topic in KSRotation and '15. Connect & Request Instructions Screen' topic in Lyracist (` HelpViewModel.cs `), providing step-by-step guidance for dual QR code setup, password management, resolution scaling, and target display selection.

Section: Singer Request Board / Patron Request Portal

[Update Details]

A new online search capability has been added directly to the patron request portal.

1. Online Song & Artist Lookup:

- When users access the Patron Request Portal via QR Code, they will see a new field labeled "Search Songs & Artists (Online)".
- Type query keywords (such as the song name or artist) to query the iTunes search index in real-time.
- Tap any suggestion from the search results list to automatically populate the "Song Title" and "Artist Name" fields in the form.

2. Connection Dependency:

- This feature runs on the patron's mobile browser, using the patron's own internet connection (via mobile data or internet-enabled Wi-Fi).
- The host system (main KJ laptop/tablet) does NOT require internet or external drives connected for this search to function.

3. Graceful Offline Fallback:

- If the patron's phone has no internet connection (e.g. no cellular signal or local-only hotspot), the search bar will gracefully show a "Search unavailable (offline)" message.
- The manual entry form remains fully functional under all network conditions.

Section: Display Projection Window / Display Monitor Selection

[Update Details]

Operators can now select a specific monitor for displaying the rotation projection window (Singer Display Window) from a dropdown list of available system screens.

1. Target Monitor Dropdown:

- Located in the "Settings" tab under a new group titled "Display Monitor".
- Displays all connected monitors with their indices, bounds, and whether they are primary.
- When selected, the display window is repositioned on that specific screen.
- If the selected monitor is unplugged or unavailable, the application falls back automatically to the secondary screen if available, or the primary screen.

2. Dynamic Window Relocation:

- Selecting a different monitor dynamically moves the active, maximized display window to the target screen immediately without needing to close and re-enable it.

3. Refresh Monitors List:

- If a new monitor is connected while the application is running, operators can click the "Refresh" button next to the dropdown to scan and reload the system's screen configuration.

Section: Rotation Management / Paused and Inactive Singer Controls

[Update Details]

Updates have been made to both the main console and the DJ web portal to manage paused and deleted (inactive) singers more effectively.

1. Paused Singers:

- Operators can pause a singer using the Pause button in the console or the remote DJ portal.
- Paused singers retain their index/position in the rotation list, but their row is grayed out.
- Paused singers cannot be marked as the current singer or next singer until they are explicitly unpaused (resumed). They are automatically skipped in the rotation.

2. Deleted/Inactive Singers:

- Clicking the Delete (trash can) button no longer deletes a singer from the database. Instead, it marks them as inactive (deleted) and moves them to the end of the rotation list.
- Inactive singers' rows are grayed out.
- When a singer is inactive, their delete button icon changes to a "↶" (Restore) button in the console and "Restore" button in the DJ portal.
- Clicking Restore reactivates the singer and moves them back to the end of the active section in the rotation.

Section: Patron Portal / Background Music & Song Requests

[Update Details]

Patrons can now request background music/song tracks to be played in between karaoke performances directly from the web portal.

1. Karaoke vs. Music Selection:

- On the Patron Request Portal, users will see a new selector under their name: "Karaoke (Sing)" and "Music (Play)".
- Selecting "Music" limits the submission to one song, hides the "Add another song" option, and marks the request category as Music.

2. DJ Incoming Requests Queue:

- On the DJ screen and the remote DJ portal web interface, incoming music requests are labeled with a green [MUSIC] badge, distinguishing them from standard purple [KARAOKE] requests.
- Standard duplicate logic is updated so a singer can have one active karaoke slot and one background music slot simultaneously.

3. Rotation List Layout:

- Accepting a music request places it in the active rotation queue.
- For music entries in the queue, the 10 round checkboxes are automatically hidden and replaced by a "♪ Background Music Track" label.
- On the DJ screen, the name, song, and artist fields of music tracks are highlighted in bold green for easier identification.

4. Projection Display Support:

- Music requests show up with a "[MUSIC]" prefix in the active list and flip tile board.
- When a music request is active/next:
 - The projection window's Vegas Marquee style shows "★ BACKGROUND MUSIC ★".
 - The Vinyl Turntable style displays "♪ NOW PLAYING".

- The Star Wars Crawl renders the entry with a "NOW PLAYING" header, displaying the song title as the primary text, followed by the artist and requester details below.

Section: Rotation Management / FlipTile Board View Removal

[Update Details]

The obsolete "Flip Tile Board" view option has been removed from all applications (including KSRotation and Lyracist).

1. Interface Clean-up:

- The "Flip Tile Board" selection option has been removed from the Projection View dropdown menus.
- All underlying split-flap character structures and visual board panels have been completely removed from the singer display screen to simplify the interface and optimize performance.

Section: Main Console / Active Venue & DJ Title Bar Integration

[Update Details]

To resolve screen space and layout wrapping issues on 1920x1080 laptop screens, the active Venue and DJ name displays have been integrated into the standard window title bar.

1. Dynamic Title Bar Display:

- The active Venue and DJ name are now visible in the title bar of the main console window, e.g. "Karaoke Singer Rotation - Venue: [VenueName] | DJ: [DjName]".

- The title updates dynamically in real-time when selecting or updating the active Venue or DJ under Settings.

2. Unwrapped Control Bar:

- The status badge has been removed from the main rotation tab toolbar, ensuring that other controls (such as checkboxes and view selectors) have maximum screen width and do not wrap layout rows.

Section: Mobile & Desktop App / Standardized Splash Screen

[Update Details]

The splash screen for the mobile and desktop versions of Karaoke Singer Rotation (KSRotation.Maui) has been redesigned and standardized using cross-platform vector assets.

1. Vector Centerpiece Logo:

- Features a clean, high-contrast microphone centerpiece surrounded by vibrant purple and orange rotating orbital paths.
- Simplified to ensure 100% compatibility with standard .NET MAUI Resizetizer compilation tools on both Android and Windows.

2. Obsidian Dark Theme Background:

- The launch background has been updated to a deep premium obsidian color (#12091A) on both platforms, providing a seamless visual transition during startup.

Section: Display Projection Window / DJ Banner and Rotation Screen Collision Priority

[Update Details]

To ensure proper projection setup, priority logic has been added when both the DJ Banner and the Rotation Display target the same screen/monitor.

1. Same-Screen Deconfliction:

- The DJ Banner will usually be displayed on the screen while everything is getting setup and the list is started.
- Once the Rotation Display is enabled (or moved) to the same screen, the DJ Banner is automatically disabled and hidden so the rotation list can display.

2. Multi-Screen Independent Operation:

- If the DJ Banner and the Rotation Display target different screens, they will both be displayed and operate simultaneously.

Section: Main Console / Front Page Music Search & Rotation Insertion

[Update Details]

Operators can now search the background music library directly from the main Karaoke dashboard, add tracks to the rotation queue as music, and configure automated playback.

1. Music Library Search Tab:

- A new "Music Library" tab has been added to the search panel on the front page of Lyracist, positioned next to the "Local Library" tab.
- Searching via the main search bar queries both the karaoke library and the background music library (tracks where IsKaraoke is false) simultaneously.

2. Insertion into Performer Rotation:

- Displays search results with Song and Artist columns.
- Click the "Add to Rotation" (+) button or double-click a song in the list to immediately insert the track into the active rotation queue as a music track (highlighted in green and labeled "♪ Background Music Track").
- Music tracks are added under the generic name "Music Request", bypassing the duplicate check and database persistence, allowing you to insert multiple independent music tracks into different positions in the queue.
- This provides a convenient mechanism for hosts to interject background music tracks into the main rotation occasionally without requiring a patron request.

3. Auto-Play & Configurable Delay:

- Added a new control row right below the queue filters on the front page containing an "Auto-Play Rotation Music" checkbox and a "Delay" selector.
- When checked, background music tracks in the rotation queue will automatically start playing after the previous singer finishes.
- The delay selector allows customizing how many seconds (0s to 30s) the system waits before starting the music track.

4. Automatic Queue Cleanup:

- Once a background music track finishes playing and is marked Done, it is automatically removed from the rotation queue rather than remaining for subsequent rounds.

Section: Display Settings & Remote DJ Board / Special Event Banners

[Update Details]

Special Event Banners have been introduced to Lyracist, KSRotation, and the remote DJ Board app. These banners share the same screen space as the standard DJ banners, allowing operators and DJs to toggle custom banners for special occasions.

1. Pre-Configured Events:

- Includes default, high-quality event graphics for "Birthday", "Wedding", "Engagement", and "Anniversary".

2. Configurable Mapping:

- In both Lyracist and KSRotation Settings panels, operators can define custom Event Names and map them to image/video files residing in the "EventBanners" directory.

3. Host Control Toggles:

- A row of radio buttons has been added directly to Lyracist's "Karaoke Page" (under DJ Banner Selection) and KSRotation's "Display" tab (under DJ & Special Event Banners).
- Selecting an event name immediately projects the mapped special event banner on the projection screen. Selecting "None" reverts the screen to the standard DJ banner.

4. Remote DJ App Integration:

- The web-based remote DJ dashboard (`dj.html`) has a new "Special Event Banners" panel.
- Polling queries the active event state from the host. Tapping any event button (or "None") updates the host display in real-time, allowing DJs to easily manage special event banners from their phone/tablet.

Section: Display Settings & Remote DJ Board / Special Event Banners Management & Styling

[Update Details]

Updates have been made to Special Event Banner management in KSRotation and the remote DJ Board app, along with high-contrast styling and optical scanning enhancements.

1. Upload, Add & Delete Special Event Controls:

- Added "Upload Banner", "Add Event", and "Delete" buttons to the Special Event Banners Mapping panel under KSRotation's Display tab.
- Uploading copies banner graphics directly to the EventBanners directory.

2. Standard Pre-Saved Event Protection & Display:

- The 4 standard pre-saved events ("Birthday", "Wedding", "Engagement", "Anniversary") are guaranteed to exist with auto-generated graphics if missing.
- Standard events are protected from accidental deletion.
- The mapping grid displays standard events as clean fixed labels, reserving drop-down list boxes and edit inputs exclusively for custom events.

3. Dual-Theme Dynamic Purple & Light Lavender UI Styling & Dark Maroon QR Code:

- Button text, borders, and RadioButton labels dynamically adapt based on theme mode: bright light lavender (#E9D5FF text, #C4B5FD border) for dark backgrounds, and rich deep purple (#6D28D9 text, #7C3AED border) for light backgrounds to ensure high legibility across all theme modes.
- The DJ QR Code foreground color has been updated from bright crimson to dark maroon/brick-red (RGB 128, 0, 32 / #800020), ensuring high optical contrast against white tiles for scanning by all mobile cameras.

4. 11+ Performance Rotation Advance:

- Clicking Finished Song in KSRotation for a performer who has completed all 10 round checkboxes now seamlessly advances the rotation, promotes queued songs, and logs performance details to the database history.

Section: Rotation Advancement & Default Event Banners Update

[Update Details]

Updates have been made to rotation sequence advancement and standard default event banners in KSRotation and Lyracist.

1. Robust 1-Click Rotation Advancement & Rollover:

- When marking a singer finished (via "Done", "✓", or remote DJ portal), rotation sequence advancement calculates the next active singer sequentially starting after the finished singer's index.
- When the last singer in the rotation finishes, rotation automatically rolls over to the top performer in 1 click.
- Rotation advances reliably even if the finishing singer was not marked as current prior to clicking "Done".

2. "Last Song" Default Event Banner:

- Added "Last Song" (` LastSong.png `) to the standard pre-saved event banners list alongside Birthday, Wedding, Engagement, and Anniversary.
- Standard pre-saved events are protected from deletion and automatically mapped to their respective graphics.

Section: Lyracist Live Trivia / In-App Help, About Tabs & Authoritative Timer Controls

[Update Details]

Dedicated Help and About systems have been added to the LyracistTrivia host application, and help topics across Lyracist and KSRotation have been updated with full trivia operations, pre-game lobby features, and timer rules.

1. LyracistTrivia "Help" and "About" Tabs:

- **Help Tab (? Help)**: Features a split-pane interactive index menu with 7 comprehensive documentation topics:

* 1.  1. Game Master Command Deck: Pack loading, 1-click pre-game lobby launch, skip countdown, auto-advance loop, manual reveals, and live response visualizer.

* 2.  2. 70:30 Pre-Game Lobby & 16:9 Banners: Widescreen 70:30 projection layout, 16:9 category banners, countdown clock, stacked Wi-Fi & Mobile Join QR codes, bottom connection bar, and scrolling ticker.

* 3.  3. Settings & Timer Override Rules: Game Master Question Answer Window timer override, 5s wrong answer elimination, 5s reveal buffer, 1,000 base points, +500 speed bonus, +10% streak multiplier, and wrong answer deductions.

* 4.  4. Mobile Buzzer Web App: Zero-install player buzzer portal (<http://<LAN-IP>:8085/trivia>), player/team registration, 4-button color buzzer, haptic feedback, real-time elimination fading, and personal ranking.

* 5.  5. Players & Teams Management: Real-time roster table, score tracking, answer streaks, accuracy stats, and instant player kick moderation.

* 6.  6. 14 Category Packs (2,100 Questions): 14 curated packs (150 questions each), SQLite database seeding, and custom JSON packs.

* 7.  7. Multi-Monitor & Shortcut Keys: Multi-monitor selection, Esc / floating close button, F11 fullscreen, and DJ banner auto-pause deconfliction.

- **About Tab (i About)**: Displays application logo, version info (v26.8.19.20), company (PAROLE Software), author (Dennis N. Maidon), copyright, and system specifications summary.

2. Authoritative Question Answer Window Timer Rule:

- The Game Master's configured Question Answer Window in Settings (``DefaultQuestionSeconds``, default: 15s) is authoritative for all live rounds.
- If the Game Master changes the timer to 10s, 20s, 25s, 30s, etc., the game engine executes every question using the Game Master's custom duration, overriding the 15s default written in the JSON files.

3. Cross-Application Help System Updates:

- ****Lyracist Help Screen (``HelpViewModel.cs``)****: Added Topics 16 ("Lyracist Live Trivia"), 17 ("Trivia Settings, Timers & Scoring Rules"), and 18 ("Trivia JSON Database Schema & Custom Pack Guide").
- ****KSRotation Help Screen (``MainWindow.xaml``)****: Updated "🎯 Trivia Night Pro" topic with 70:30 pre-game lobby, 16:9 category banners, 14 categories (2,100 questions), timer override rules, and custom JSON packs.

Section: Trivia Database Creator & Pack Studio (``TriviaDbCreator.exe``)

[Update Details]

A dedicated standalone question authoring and category management studio (``TriviaDbCreator.exe``) has been introduced to the Lyracist suite.

1. Fluent UI / MVVM Architecture:

- Built with WPF .NET 10, CommunityToolkit.Mvvm, and WPF-UI (Fluent Dark Theme / Mica styling).
- Real-time dirty state tracking, auto-save validation, and clean navigation.

2. Pack & Question Management Features:

- ****Left Sidebar Pack Explorer****: Auto-discovers and navigates all JSON question packs in ``TriviaData/packs/`` with search filtering and question count indicators.

- **Pack Metadata Editor**: Configure Pack Title, ID slug, Category, and Theme Description.
- **Center Panel Questions Table**: Filterable table listing Question #, ID, Prompt, Difficulty badge (` Easy` / ` Medium` / ` Hard`), and Correct Option indicator (` ▲ A` , ` ◆ B` , ` ● C` , ` ■ D`).
- **Right Panel Question Editor**: Edit question prompt, 4 distinct multiple-choice options (A, B, C, D), radio selector for correct answer, difficulty rating, and reveal explanation note.
- **1-Click Option Balancing**: Automatically shuffles option positions across all questions in a pack to guarantee an even ~25% distribution across choices A, B, C, and D without altering question accuracy.
- **16:9 Banner Studio**: Generates matching high-resolution 16:9 Category Announcement Banners (` TriviaData/Banners/{pack}.png`) with one click.
- **Direct SQLite Seeding**: 1-click database synchronization updating `TriviaData/trivia.db` .

3. In-App Help & Documentation System (` ? Help & Instructions`):

- Built-in split-pane documentation browser covering 8 in-depth topics:
 - * **1. Quick Start & Overview**: General workflow and 3-column workspace orientation.
 - * **2. Creating & Managing Databases**: How to add, configure, rename, and delete category packs.
 - * **3. Adding, Editing & Duplicating Questions**: Multi-choice option inputs, radio toggles for correct answer, difficulty ratings, and reveal explanations.
 - * **4. Balancing Answer Distributions (25% Rule)**: Real-time distribution analysis and 1-click option shuffling.
 - * **5. 16:9 Category Announcement Banners**: Widescreen banner rendering into `TriviaData/Banners/` .
 - * **6. SQLite Database Sync & Live Hosting**: Cross-application sharing with `LyracistTrivia` , `Lyracist` , and `KSRotation` .
 - * **7. Importing & Exporting Databases**: Backup, restore, and pack exchange.

* **8. JSON Database Format & Schema Reference***: Syntax rules, sample JSON code, and field specifications.

- **  About Tab***: Displays official brand logo, version, company, author, copyright, and technology stack.

Section: Full Trivia System Alignment & Screen Parity Across Solution Applications
(Lyracist, KSRotation, Lyracist.Trivia)

[Update Details]

Complete visual, architectural, and operational feature parity has been implemented across the embedded Trivia engines in Lyracist and KSRotation to match the standalone Lyracist.Trivia Game Master application. Furthermore, strict screen display prioritization and automatic pause deconfliction ensure that live Trivia seamlessly integrates around primary karaoke singer performance and rotation workflows.

1. 16:9 Big Screen Projection Window (` TriviaDisplayWindow`) in Lyracist & KSRotation:

- **Unified 70:30 Pre-Game Lobby Layout**:

* **70% Left Hero Column***: Full widescreen showcase displaying high-resolution 16:9 Category Announcement Banners (` TriviaData/Banners/{pack}.png`) or real-time generated collage banners for multi-pack mixed rounds (` TriviaBannerGenerator`). Includes category theme title, subtitle, and ambient illumination.

* **30% Right Onboarding Stack**:

1. **  Pre-Game Countdown Clock***: Digital countdown timer displaying time remaining until round start.

2. **  1. Connect to Wi-Fi QR Card***: Scan-to-connect Wi-Fi QR code with venue SSID and WPA password for instant patron phone connection.

3. **  2. Join Trivia Game QR Card***: Scan-to-join mobile buzzer QR code pointing to `http://<LAN-IP>:8085/trivia` .

* **Bottom Connection & Copyright Bar**: Displays mobile buzzer URL, company copyright (` © 2026 PAROLE Software `), and app branding.

* **Top Scrolling Marquee Bar**: Continuous horizontal ticker cycling venue announcements, host branding, game rules, and top scorer leaderboards with crown indicators.

- **Active Gameplay & Question Presentation**:

* Scaled question prompt cards with automatic Viewbox text fitting.

* 2x2 stylized answer option cards (▲ Red/Purple, ◆ Cyan/Blue, ● Amber, ■ Rose/Green) with real-time wrong option elimination fading and correct answer highlights upon reveal.

- **Post-Game Celebration & Intermission**:

* Winner trophy fanfare with individual champion score or winning team roster breakdown.

* Live intermission timer countdown banner indicating when the next round will auto-start.

2. 3-Column Game Master Console Deck (Lyracist ` TriviaPage.xaml ` & KSRotation ` MainWindow.xaml `):

- **Header Bar**: Active venue indicator, live connected player count badge, target monitor projection dropdown, and 1-click "  Open TV Display" / "  Open Big Screen" button.

- **Left Column (Game Master Controls)**:

* Category Pack Checklist (check 2+ to mix categories).

* Questions Per Game preset selector (5, 10, 15, 20, 25, 50, 100 questions).

* Auto-Run Game switch (15s answer, 5s fade, 5s reveal).

* 1-Click ` "  Launch Pre-Game Countdown" ` and ` "  Start Game Now (Skip Countdown)" ` .

* Live Question Controls (Start Question, Pause/Resume, Lock & Reveal Answer, Next Question, Reset Session).

* Mobile Web Server status and direct join link.

- **Center Column (Live Round & Answer Distribution Visualizer)**:

- * Active Round header and ticking countdown timer.

- * Current Question prompt card.

- * Real-time **Patron Answer Distribution Visualizer** displaying live vote counts, percentages, and progress bars for options A, B, C, and D.

- * 2x2 Answer Option previews.

- **Right Column (Live Player Leaderboard & Moderation)**:

- * Live player roster sorted by total score.

- * Player name, registered team name, and current score.

- * Individual player kick action (`X`) for removing disruptive participants.

3. Dedicated Separate Trivia Settings (Lyracist `TriviaSettingsPage.xaml` & KSRotation `MainWindow.xaml`):

- **Question Timers & Flow**: Time limit (s), Warning countdown (s), Wrong option elimination fade (s), Post-reveal delay (s), and Auto-advance toggle.

- **Scoring & Multipliers**: Base points per question (1,000 pts), Wrong answer deduction (0 pts default), Speed bonus toggle & maximum (+500 pts), Streak bonus toggle & multiplier (+10%/streak).

- **Venue & Host Customization**: Active venue name, Host/DJ name, and customizable Pre-Game Welcome Banner template supporting `{venue}` and `{dj}` replacement tokens.

- **Intermission & Pre-Game Controls**: Auto-start next game toggle & delay in minutes, Pre-game countdown delay in minutes & auto-start toggle upon timer expiry.

- **Local Network & Wi-Fi Access**: Server port (8085 default), Wi-Fi SSID, Wi-Fi password, and 1-click ` "Auto-Detect Wi-Fi" ` tool.

- **Target Projection Monitor**: Selectable display monitor with true physical DPI awareness.

- **Save & Reset**: 1-click Save Settings and Reset to Defaults with animated status toasts.

4. Karaoke & Rotation Screen Priority / Automatic Pause Deconfliction:

- In both Lyracist and KSRotation, Trivia operates as a secondary entertainment module that automatically respects and yields to primary karaoke and singer rotation functions:

* **Priority 1***: Karaoke Lyrics Projection Window during song playback.

* **Priority 2***: Singer Rotation Queue Display between songs.

* **Priority 3***: DJ / Venue Announcement Banners.

* **Priority 4***: Lyracist Live Trivia Projection.

- When a singer performs or the rotation screen is cast, the Trivia engine automatically pauses with a clear pause status (` "Karaoke Performance" ` or ` "Rotation Screen Active" `) and hides or yields the display.

- Once song playback or rotation display concludes, Trivia automatically resumes seamlessly.

Section: Live Trivia Scoring / Tiered Option Value Multiplier (100% / 70% / 40%)

[Update Details]

A dynamic tiered scoring system has been integrated into Lyracist, KSRotation, and Lyracist.Trivia, tying point awards directly to the visual option elimination phases during live question play.

1. Progressive Option Elimination & Value Tiers:

- **4 Options Visible (0 Eliminated)***: 100% of Base Points (1,000 pts default). Rewarding early, confident buzz-ins before any wrong answers fade.

- **3 Options Visible (1 Eliminated)***: 70% of Base Points (700 pts default). Occurs when the first wrong answer fades at the 2/3 countdown interval.

- **2 Options Visible (2 Eliminated / 50-50)***: 40% of Base Points (400 pts default). Occurs when the second wrong answer fades at the 1/3 countdown interval.

- ****Expected Value Balance****: Discourages patrons from blindly waiting for the 50/50, ensuring early knowledge is awarded a decisive 2.5× advantage (1,000 vs 400 pts) while casual patrons can still score points.

2. Mobile Buzzer Web App (` trivia.html `):

- Real-time Point Multiplier Pill (` #value-multiplier-pill `) prominently displayed above the buzzer grid.

- Dynamically updates as options fade: ` ⚡ 100% VALUE (1,000 pts) ` , ` ⚡ 70% VALUE (700 pts) ` , ` ⚡ 40% VALUE (400 pts) ` .

- Wrong options fade and disable on the phone in real time as they fade on the big screen.

3. Trivia Settings & Configuration:

- Toggle checkbox: "Tiered Option Value (4=100%, 3=70%, 2=40%)"

- Customizable percentage fields for 4 Opt (%), 3 Opt (%), and 2 Opt (%) across Lyracist (` TriviaSettingsPage.xaml `), KSRotation (` MainWindow.xaml `), and Lyracist.Trivia (` MainWindow.xaml `).

Section: Database Manager / Catalog Book Exporter & Failed Artist List

[Update Details]

Updates have been made to the Catalog Book Exporter formats and the Slow Metadata Extractor.

1. Supported Formats and Sorting:

- The Catalog Book Exporter now supports exporting the catalog as a plain text (.txt) file.
- The format selection dropdown has been sorted in the following order: PDF Document (.pdf), Text File (.txt), and Word Document (.docx).

2. Metadata Scan Failed Artist Export:

- When running a Slow Metadata Scan for songs with unknown artists, any files that were unable to be resolved and updated will be listed in the "Failed Artist List".
- Upon completion or cancellation of the scan, a plain text file listing all unresolved file paths is automatically exported to the "Releases" folder.

Section: Display Projection Window / Casting Support

[Update Details]

Operators can now select different casting targets for the Singer Rotation Billboard screen in both Lyracast and KSRotation applications.

1. Casting Target Dropdown:

- Located in the "Settings" tab under "Displays & Projection" (or "Display & Projection" in KSRotation).
- Dropdown options include Monitor, Miracast, Chromecast, Browser Cast, AirPlay, and Wireless HDMI.
- When a casting target is selected, the application (Lyracast or KSRotation) initializes the appropriate casting sender/server in the background.

2. Off-Screen Rendering:

- Selecting a casting target (such as Chromecast or Browser Cast) creates and renders the Rotation Window / Singer Display Window in an off-screen canvas.
- This ensures the KJ laptop screen is not cluttered while casting frames.

3. Browser Cast Server:

- The self-hosted Browser Cast option spins up a local web server (<http://localhost:8080/rotation/>) to allow any browser on the local network to access the casting stream.

Section: Performer Profiles, Song Tagging & Singer Avatars

[Update Details]

A comprehensive performer profile system and song tagging support have been integrated into Lyracist.

1. Performer Profiles & Login:

- Patrons accessing the Patron Request Portal can log in or register using their Performer Name and a 4-digit PIN.
- Profile settings include email registration, custom vocal range specification (e.g. Tenor, Soprano), and custom title assignment

2. Singer Avatars & Custom Uploads:

- Performers can customize their profile avatar using a dropdown selector.
- Supports default silhouette, Gravatar logo generation (via email MD5 hash), and direct mobile photo/selfie uploads (capped at
- Avatars are displayed as circular icons in the rotation board on both the mobile Patron Portal and host WPF Singer Queue List

3. Database Tagging:

- Songs can now be tagged with comma-separated categories to organize, filter, and challenge performers.

Section: Integration Settings / Party Tyme Removal

[Update Details]

The Party Tyme karaoke streaming integration has been removed from the application.

1. Removal of Streaming Tab:

- The "Party Tyme Online" tab has been removed from the main Karaoke dashboard view.
- The "Party Tyme (Streaming)" tab has been removed from the Special Occasions Music search dialog.

2. Settings Panel Clean-up:

- The subscription fields (Client ID and Client Secret) for Party Tyme have been removed from the Settings panel under APIs & Integrations.

3. Queue & Badge Updates:

- The "Party Tyme" filter checkbox and visual badge triggers have been removed from the singer rotation queue list.

Section: Database Manager / Catalog Book Exporter

[Update Details]

Operators can now export the entire song database (Karaoke or Music) into Word (.docx) or PDF format directly from the Lyracist

1. Exporter Panel:

- Located in the right pane of the Lyracist Database Manager under "Catalog Book Generator".
- Provides two dropdown selectors:
 - Catalog: Choose between the "Karaoke Catalog" (tracks marked as karaoke) and the "Music Catalog" (standard audio tracks).
 - Format: Choose between "Word Document (.docx)" and "PDF Document (.pdf)".

2. Generated Fields & Formatting:

- The generated books are structured as clean, professionally styled tables showing:
 - Artist: Alphabetically ordered performer names.
 - Title: Song titles.
 - Length: Formatted track durations (MM:SS).
 - Genre: The embedded audio genre (automatically populated during metadata scans).

3. Pagination & Output:

- The document is fully paginated, displaying "Page X" or "Page X of Y" in the bottom footer.
- For Word (.docx) books, a repeating header row is defined so column titles automatically reappear on the top of each page.
- When generation completes, Lyracist automatically launches the default system viewer to open and print the document, which